

# Day 18 - Lecture 18.13: Scatter Plots

## (`plt.scatter`)

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### 1. Overview & Purpose

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A **Scatter Plot** displays values for two numerical variables as points on a 2D plane. Unlike line plots, **no line segments connect the points**. Each point represents a single observation or subject.

#### Primary Use Cases:

- **Correlation Analysis:** Is there a positive, negative, or zero correlation between  $X$  and  $Y$ ?
  - **Non-Linear Relationships:** Detecting curvilinear or exponential trends.
  - **Clustering:** Detecting naturally occurring groupings in feature space.
  - **Outlier Spotting:** Immediately locating isolated points far away from the main cluster.
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### 2. Syntax & Basic Parameters

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```
plt.scatter(  
    x,                # Continuous independent variable  
    y,                # Continuous dependent variable  
    s=None,           # Marker size (scalar or array)  
    c=None,           # Marker color (scalar or array of values)  
    marker='o',        # Shape: 'o', 's', '^', 'x', 'd'  
    alpha=None,        # Transparency (0.0 to 1.0)  
    linewidths=None,   # Border outline width  
    edgecolors=None    # Border outline color  
)
```

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### 3. Real-World Case Study: Age vs. Blood Pressure

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Medical analysis of 10 patients studying whether systolic blood pressure increases monotonically with age:

```
import matplotlib.pyplot as plt

age = [22, 25, 30, 35, 40, 45, 50, 55, 60, 65]
blood_pressure = [110, 115, 120, 122, 125, 130, 135, 123, 145, 150]

plt.figure(figsize=(8, 5))
plt.scatter(age, blood_pressure, color="crimson", s=80, edgecolor="black", alpha=0.8)

plt.title("Medical Analysis: Age vs Blood Pressure", fontsize=14, fontweight="bold")
plt.xlabel("Age (years)", fontsize=12)
plt.ylabel("Blood Pressure (mmHg)", fontsize=12)
plt.grid(True, linestyle=":", alpha=0.6)
plt.show()
```

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### 4. Key Takeaways

- The Pearson correlation coefficient  $r \approx 0.96$  indicates a strong positive linear relationship between age and blood pressure.
- Notice patient at Age 55 with BP 123 mmHg dips below the trendline—scatter plots make individual variances instantly visible.